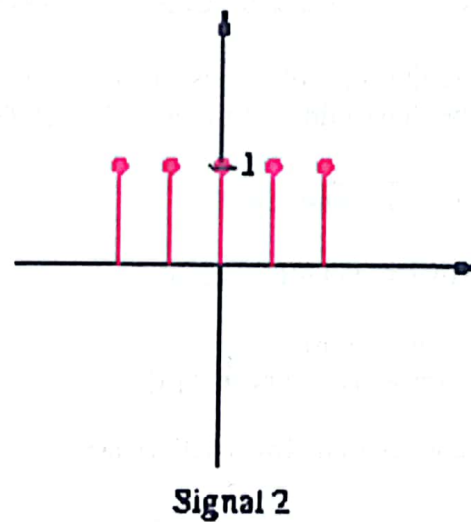
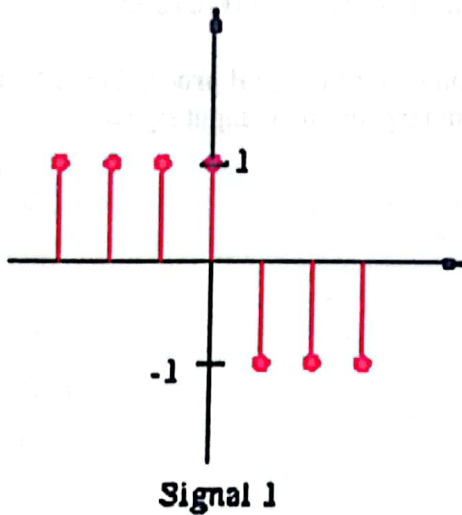


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Lecture 5

DT Convolution



Discrete-Time Convolution (DT Convolution)

Discrete-time convolution is a fundamental operation in **digital signal processing (DSP)**. It describes how a **linear time-invariant (LTI)** system responds to an input signal.

1- Definition

Given two discrete-time signals:

- **Input:** $x[n]$
- **Impulse response:** $h[n]$

Their convolution $y[n]$ is defined as:

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k]$$

Convolution tells us:

How the system with impulse response $h[n]$ reacts to the input $x[n]$.

It is like sliding the signal $h[n]$ over $x[n]$, multiplying, and summing.

Step-by-Step Procedure

To compute $y[n] = x[n] * h[n]$:

- ✓ 1. Flip one of the signals $\rightarrow$ compute $h[-k]$
- ✓ 2. Shift it by $n \rightarrow$ compute $h[n - k]$
- ✓ 3. Multiply: $x[k] \cdot h[n - k]$
- ✓ 4. Sum for all values of k

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Example 1:

Let:

$$x[n] = [1, 2, 1]$$

$$h[n] = [1, 1]$$

Compute:

$$y[n] = x[n] * h[n]$$

input
response

Signal Length

- Length of $x = 3$
- Length of $h = 2$
- Length of $y = (3+2)-1=4$

So $y[n]$ will have ~~five~~ ^{four} values: $y[0]$ to $y[3]$.

For $n=0$ $\rightarrow$

$$y[0] = 1 \cdot 1 = 1$$

For $n=1$ $\rightarrow$

$$y[1] = 2 \cdot 1 + 1 \cdot 1 = 3$$

For $n=2$ $\rightarrow$

$$y[2] = 1 \cdot 1 + 2 \cdot 1 = 3$$

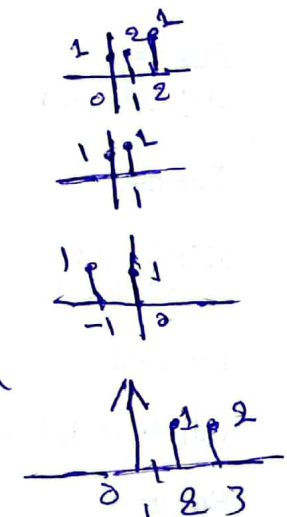
For $n=3$ $\rightarrow$

$$y[3] = 1 \cdot 1 = 1$$

Therefore

$$y[n] = [1, 3, 3, 1]$$

Flip
Shift
Multiply
Summation



Example 2:

Let

$$x[n] = [2, 1, -1]$$

$$h[n] = [1, 0, 1]$$

$$y[n] = x[n] * h[n]$$

1-Signal Length

- Length of $x = 3$
- Length of $h = 3$
- Length of $y = 3+3-1=5$

So $y[n]$ will have five values: $y[0]$ to $y[4]$.

For $n=0$

Only one overlap:

$$y[0] = 2 \cdot 1 = 2$$

For $n=1$

$$y[1] = 2 \cdot 0 + 1 \cdot 1 = 1$$

For $n=2$

$$y[2] = 2 \cdot 1 + 1 \cdot 0 + (-1) \cdot 1$$

$$y[2] = 2 + 0 - 1 = 1$$

For $n=3$

$$y[3] = 1 \cdot 1 + (-1) \cdot 0 = 1$$

For $n=4$

$$y[4] = (-1) \cdot 1 = -1$$

Final Result

$$y[n] = [2, 1, 1, 1, -1]$$

Example 3

Let

$$x[n] = [1, 2, 1, -1]$$

$$h[n] = [1, 0, 1]$$

$$y[n] = x[n] * h[n]$$

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Signal Lengths

- Length of $x = 4$
- Length of $h = 3$
- Length of output:

$$L_y = 4 + 3 - 1 = 6$$

So $y[n]$ has 6 values: $y[0]$ to $y[5]$

For $n=0$

$$y[0] = 1 \cdot 1 = 1$$

For $n=1$

$$y[1] = 1 \cdot 0 + 2 \cdot 1 = 2$$

For $n=2$

$$y[2] = 1 \cdot 1 + 2 \cdot 0 + 1 \cdot 1$$

$$y[2] = 1 + 0 + 1 = 2$$

For $n=3$

$$y[3] = 2 \cdot 1 + 1 \cdot 0 + (-1) \cdot 1$$

$$y[3] = 2 + 0 - 1 = 1$$

For $n=4$

$$y[4] = 1 \cdot 1 + (-1) \cdot 0 = 1$$

For $n=5$

$$y[5] = (-1) \cdot 1 = -1$$

output

$$y[n] = [1, 2, 2, 1, 1, -1]$$



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Thankyou